

PSGAN: Pose and Expression Robust Spatial-Aware GAN for Customizable Makeup Transfer

Wentao Jiang¹, Si Liu¹, Chen Gao^{2,4}, Jie Cao^{3,4}, Ran He^{3,4}, Jiashi Feng⁵, Shuicheng Yan⁶

¹School of Computer Science and Engineering, Beihang University

²Institute of Information Engineering, Chinese Academy of Sciences

³Institute of Automation, Chinese Academy of Sciences

⁴University of Chinese Academy of Sciences ⁵National University of Singapore ⁶YITU Tech

{jiangwentao, liusi}@buaa.edu.cn, gaochen@iie.ac.cn, jie.cao@cripac.ia.ac.cn

rhe@nlpr.ia.ac.cn, elefjia@nus.edu.sg, shuicheng.yan@yitu-inc.com

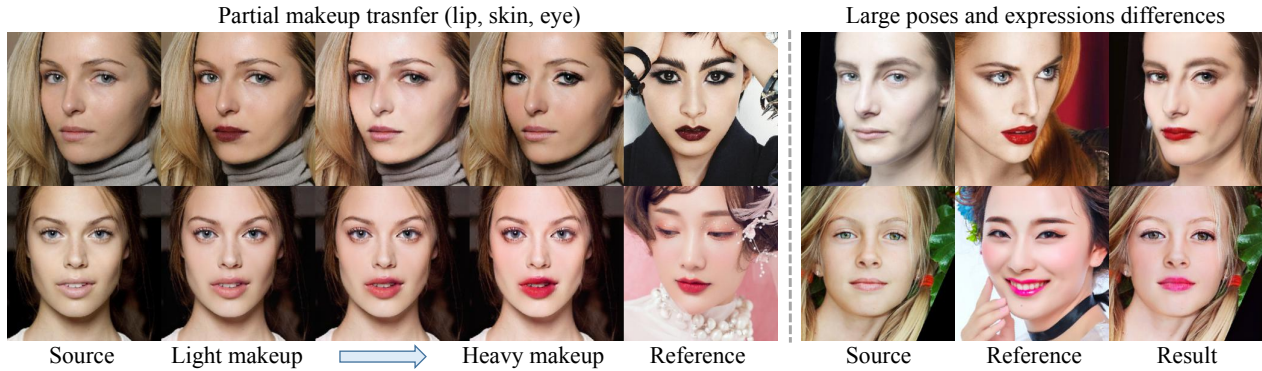


Figure 1. Our model allows users to control both the shade of makeup and facial parts to transfer. The first row on the left shows the results of only transferring partial makeup style from the reference. The second row shows the results with controllable shades. Moreover, our method can perform makeup transfer between images that have different poses and expressions, as shown on the right part of the figure. Best viewed in color.

Abstract

In this paper, we address the makeup transfer task, which aims to transfer the makeup from a reference image to a source image. Existing methods have achieved promising progress in constrained scenarios, but transferring between images with large pose and expression differences is still challenging. Besides, they cannot realize customizable transfer that allows a controllable shade of makeup or specifies the part to transfer, which limits their applications. To address these issues, we propose Pose and expression robust Spatial-aware GAN (PSGAN). It first utilizes Makeup Distill Network to disentangle the makeup of the reference image as two spatial-aware makeup matrices. Then, Attentive Makeup Morphing module is introduced to specify how the makeup of a pixel in the source image is morphed from the reference image. With the makeup matrices and the source image, Makeup Apply Network is used to perform makeup transfer. Our PSGAN not only achieves state-of-the-art results even when large pose and expression differences exist

but also is able to perform partial and shade-controllable makeup transfer. We also collected a dataset containing facial images with various poses and expressions for evaluations.

1. Introduction

We explore the makeup transfer task, which aims to transfer the makeup from an arbitrary reference image to a source image. It is widely demanded in many popular portrait beautifying applications. Most existing makeup transfer methods [17, 3, 2, 10] are based on Generative Adversarial Networks (GANs) [9]. They generally use face parsing maps and/or facial landmarks as a preprocessing step to facilitate the subsequent processing and adopt the framework of CycleGAN [31] which is trained on unpaired sets of images, i.e., non-makeup images and with-makeup images.

However, existing methods mainly have two limitations. Firstly, they only work well on frontal facial images with neutral expression since they lack a specially designed mod-

PSGAN: 姿态与表情鲁棒的空间感知生成对抗网络, 用于定制化妆容迁移

江 Wentao¹, 刘 Si¹, 高 Chen^{2,4}, 曹 Jie^{3,4}, 何 Ran^{3,4}, 冯 Jiashi⁵, 严 Shuicheng⁶ ¹北京航空航天大学计算机学院 ²中国科学院信息工程研究所 ³中国科学院自动化研究所 ⁴中国科学院大学 ⁵新加坡国立大学 ⁶依图科技

{jiangwentao, liusi}@buaa.edu.cn, gaochen@iie.ac.cn, jie.cao@cripac.ia.ac.cn
rhe@nlpr.ia.ac.cn, elefjia@nus.edu.sg, shuicheng.yan@yitu-inc.com

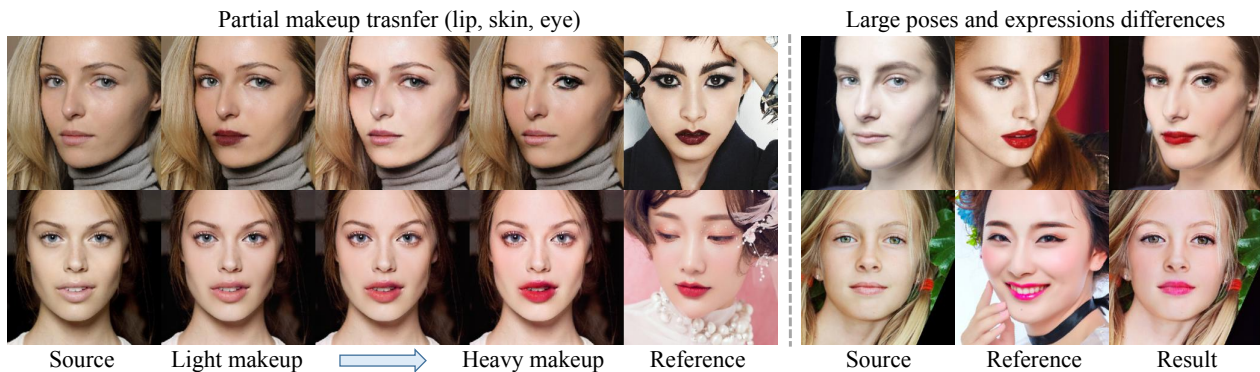


图1. 我们的方法允许用户控制化妆的颜色深浅以及面部部位的转移。左上一行展示了仅从参考图像中部分转移妆容风格的结果。第二行显示了可调节颜色深浅的效果。此外, 我们的方法还可以在姿态和表情不同的图像之间进行化妆转移, 如图右侧所示。彩色观看效果更佳。

摘要

In this paper, we address the makeup transfer task, which aims to transfer the makeup from a reference image to a source image. Existing methods have achieved promising progress in constrained scenarios, but transferring between images with large pose and expression differences is still challenging. Besides, they cannot realize customizable transfer that allows a controllable shade of makeup or specifies the part to transfer, which limits their applications. To address these issues, we propose Pose and expression robust Spatial-aware GAN (PSGAN). It first utilizes Makeup Distill Network to disentangle the makeup of the reference image as two spatial-aware makeup matrices. Then, Attentive Makeup Morphing module is introduced to specify how the makeup of a pixel in the source image is morphed from the reference image. With the makeup matrices and the source image, Makeup Apply Network is used to perform makeup transfer. Our PSGAN not only achieves state-of-the-art results even when large pose and expression differences exist

but also is able to perform partial and shade-controllable makeup transfer. We also collected a dataset containing facial images with various poses and expressions for evaluations.

1. 引言

我们研究了妆容迁移任务, 该任务旨在将任意参考图像中的妆容转移到目标图像上。这一需求在许多流行的肖像美化应用中广泛存在。大多数现有的妆容迁移方法[17,3,2,10]都是基于生成对抗网络 (GAN) [9]。它们通常使用面部解析图和/或面部特征点作为预处理步骤, 以方便后续处理, 并采用CycleGAN[31]框架进行训练, 该框架在未配对的图像集上进行训练, 即无妆图像和有妆图像。

然而, 现有方法主要存在两个限制。首先, 它们仅在正面且表情自然的面部图像上表现良好, 因为缺乏专门设计的模块来应对其他情况。

ule to handle the misalignment of images and overfit on frontal images. While in practical applications, an ideal method should be *pose and expression robust*, which is able to generate high-quality results even if source images and reference images show different poses and expressions. Secondly, the existing methods cannot perform customizable makeup transfer since they encode makeup styles into low dimension vectors which lose the spatial information. An ideal makeup transfer method need be capable of realizing *partial* and *shade-controllable* makeup transfer. Partial transfer indicates transferring the makeup of specified facial regions separately, e.g., eye shadows or lipstick. Shade-controllable transfer means the shade of the transferred makeup can be controllable from light to heavy.

To solve these challenges, we propose a novel Pose and expression robust Spatial-aware GAN, which consists of a Makeup Distill Network (MDNet), an Attentive Makeup Morphing (AMM) module and a Makeup Apply Network (MANet). Different from the previous approaches that simply input two images into the network or recombine makeup latent code and identity latent code to perform transfer, we design PSGAN to transfer makeup through scaling and shifting the feature map for only once, inspired by style transfer methods [13, 5]. Comparing with general style transfer, makeup transfer is more difficult since the human perception system is very sensitive to the artifacts on faces. Also, makeup styles contain subtle details in each facial region instead of general styles. To this end, we propose MDNet to disentangle the makeup from the reference image into two makeup matrices, i.e., the coefficient matrix γ and bias matrix β which both have the same spatial dimensions with visual features. These matrices embed the makeup information and serve as the shifting and scaling parameters. Then, γ and β are morphed and adapted to the source image by the AMM module which calculates an attentive matrix A to produce adapted makeup matrices γ' and β' . The AMM module utilizes the face parsing maps and facial landmarks to build the pixel-wise correspondences between source images and reference images, which solves the misalignment of faces. Finally, the proposed MANet conducts makeup transfer through applying pixel-wise multiplication and addition on visual features using γ' and β' .

Since the makeup style has been distilled in a spatial-aware way, *partial transfer* can be realized by applying masks pixel-wisely according to the face parsing results. For example, in the top left panel of Figure 1, the lip gloss, skin and eye shadow can be individually transferred from the reference image to the source image. *Shade-controllable transfer* can be realized through multiplying the weights of makeup matrices by coefficient within $[0, 1]$. As shown in the bottom left panel of Figure 1, where the makeup shade is increasingly heavier. Moreover, the novel AMM module effectively assists the generating of *pose and*

expression robust results, as shown in the right part of Figure 1. We also directly apply transfer to every frame of facial videos and still get nice and consistent results. With the three novel components, PSGAN satisfies the requirements we pose for an ideal customizable makeup transfer method.

We make the following contributions in this paper:

- To our best knowledge, PSGAN is the first to simultaneously realize partial, shade-controllable, and pose/expression robust makeup transfer, which facilitates the applications in the real-world environment.
- A MDNet is introduced to disentangle the makeup from the reference image as two makeup matrices. The spatial-aware makeup matrices enable the flexible partial and shade-controllable transfer.
- An AMM module that adaptively morphs the makeup matrices to source images is proposed, which enables pose and expression robust transfer.
- A new Makeup-Wild dataset containing images with diverse poses and expressions is collected for better evaluations.

2. Related Work

2.1. Makeup Transfer

Makeup transfer has been studied a lot these years [25, 11, 16, 20, 19, 1]. BeautyGAN [17] first proposed a GAN framework with dual input and output for makeup transfer and removal simultaneously. They also introduced a makeup loss that matches the color histogram in different parts of faces for instance-level makeup transfer. BeautyGlow [3] proposed a similar idea on the Glow framework and decomposed makeup component and non-makeup component. PairedCycleGAN [2] employed an additional discriminator to guide makeup transfer using pseudo transferred images generated by warping the reference face to the source face. LADN [10] leveraged additional multiple overlapping local discriminators for dramatic makeup transfer. However, the above approaches often fail on transferring in-the-wild images and cannot adjust transfer precisely and partially, which limits their applications, such as the makeup transfer in videos.

2.2. Style Transfer

Style transfer has been investigated extensively [7, 6, 14, 21, 24]. [8] proposed to derive image representations from CNN, which can be separated and recombined to synthesize images. Some methods are developed to solve the fast style transfer problem. [5] found the vital role of normalization in style transfer networks and achieved fast style transfer by the conditional instance normalization. While their methods

为了处理图像错位和正面图像过拟合的问题。而在实际应用中，理想的方法应该是

pose and expression robust，即使源图和参考图表现出不同的姿态和表情，也能生成高质量的结果。其次，现有的方法无法实现可定制的妆容迁移，因为它们将妆容风格编码为低维向量，从而丢失了空间信息。理想的妆容迁移方法需要能够实现*partial*和*shade-controllable*妆容迁移。部分迁移指的是分别转移指定面部区域的妆容，例如眼影或口红。可控阴影迁移意味着可以控制转移到妆容深浅程度，从轻到重。

为了解决这些挑战，我们提出了一种新颖的姿态和表情鲁棒的空间感知生成对抗网络（PSGAN），该网络由化妆蒸馏网络（MDNet）、注意力化妆变形模块（AMM）以及化妆应用网络（MANet）组成。与以往的方法简单地将两张图像输入到网络中或将妆容潜在代码和身份潜在代码重新组合以进行转换不同，我们设计了PSGAN通过缩放和移动特征图仅一次来实现妆容的转移，这受到了风格迁移方法[13, 5]的启发。相比一般的风格迁移，化妆迁移更加困难，因为人类感知系统对面部的伪影非常敏感。此外，化妆风格在每个面部区域包含细微的细节，而不仅仅是通用风格。为此，我们提出了MDNet将参考图像中的妆容分解为两个妆容矩阵，即系数矩阵 γ 和偏置矩阵 β ，这两个矩阵都具有与视觉特征相同的空间维度。这些矩阵嵌入了化妆信息，并作为缩放和平移参数使用。然后，通过AMM模块计算注意力矩阵 A 来生成适应源图像的妆容矩阵 γ' 和 β' 。AMM模块利用面部解析图和面部关键点在源图像与参考图像之间建立像素级对应关系，从而解决了面部对齐的问题。最后，所提出的MANet通过使用 γ' 和 β' 进行逐像素的乘法和加法操作来实现化妆迁移。

由于妆容风格以空间感知的方式被提炼出来，可以通过根据面部解析结果逐像素应用掩码来实现*partial transfer*。例如，在图1的左上部分，唇彩、肤色和眼影可以从参考图像分别转移到源图像中。*Shade-controllable transfer*可以通过将化妆矩阵的权重与[0, 1]范围内的系数相乘来实现，如图1底部左侧所示，其中妆容逐渐加重。此外，新型AMM模块有效辅助了*pose and*的生成。

expression robust 如图1右侧所示为实验结果。我们还将迁移技术直接应用于面部视频的每一帧，依然获得了良好且一致的结果。凭借这三个新颖组件，PSGAN满足了对理想可定制化妆容迁移方法的要求。

本文的贡献如下：

- 据我们所知，PSGAN 是首个同时实现部分化妆转移、可控制阴影以及姿态/表情鲁棒性的方法，这促进了其在真实世界环境中的应用。
- 引入了MDNet来将妆容从参考图像中分离为两个妆容矩阵。这些具有空间感知的妆容矩阵能够实现灵活的部分和阴影可控转移。
- 提出了一种自适应变换 Makeup Matrices (MM) 以适配源图像的AMM模块，该模块能够实现姿态和表情鲁棒传输。
- 一个新的包含多样姿态和表情的 Makeup-Wild 数据集被收集起来以进行更好的评估。

2. 相关工作

2.1. 化妆转移技术

近年来，妆容迁移的研究取得了显著进展[25, 11, 16, 20, 19, 1]。BeautyGAN [17] 首次提出了一种用于同时实现妆容迁移和去除的双输入输出生成对抗网络框架，并引入了匹配面部不同部位颜色直方图的妆容损失函数，以实现实例级别的妆容迁移。Beauty-Glow [3] 在 Glow 框架上提出了类似的概念，并将妆容成分与非妆容成分进行了分解。PairedCycleGAN [2] 采用了一个额外的判别器来引导使用参考面部变形到源面部生成伪转移图像进行妆容迁移。LADN [10] 利用多个重叠局部判别器实现了戏剧性的妆容迁移。然而，上述方法在处理野外图像时往往效果不佳，并且无法精确和部分地调整迁移，这限制了它们的应用范围，例如视频中的妆容迁移。

2.2. 风格转换

风格转换的研究已经非常广泛[7, 6, 14, 21, 24]。文献[8]提出从卷积神经网络（CNN）中导出图像表示，这些表示可以被分离和重新组合以合成新的图像。一些方法被开发出来用于解决快速风格转换的问题。文献[5]发现了归一化在风格转换网络中的关键作用，并通过条件实例归一化实现了快速的风格转换。尽管他们的方法

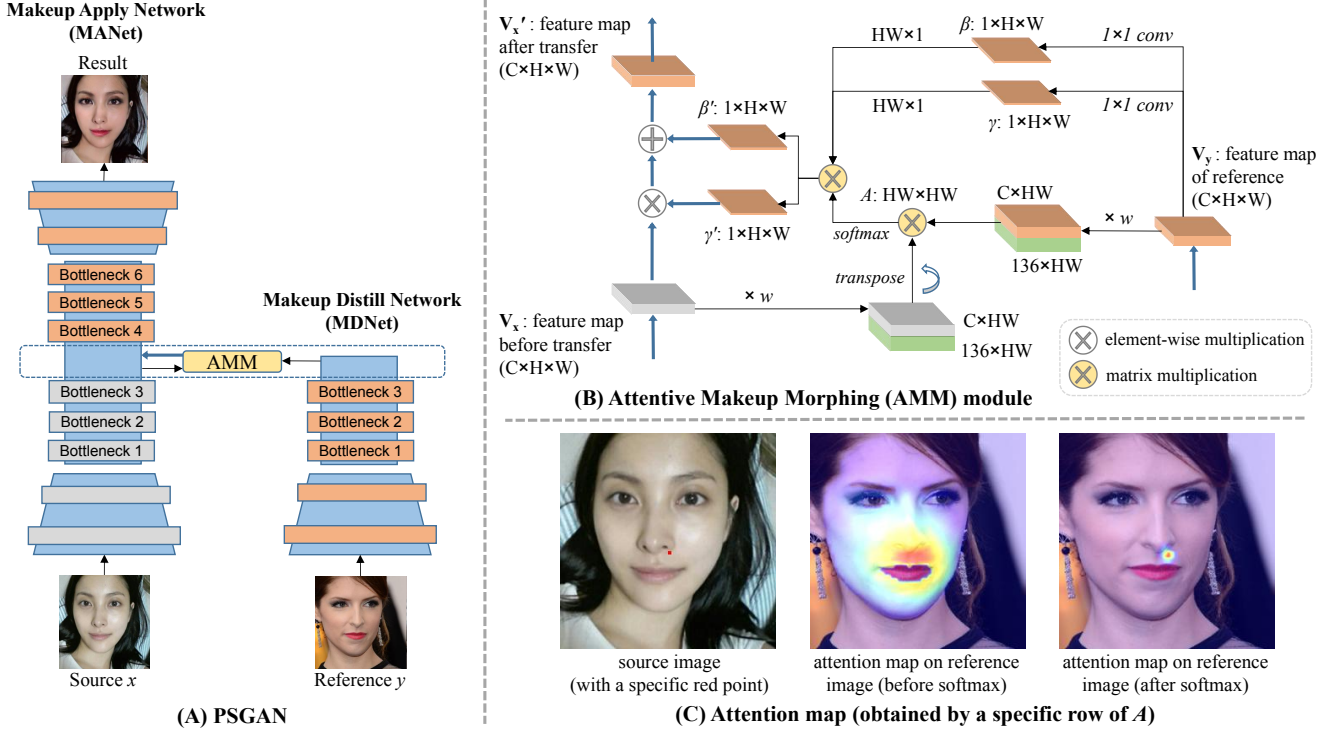


Figure 2. (A) Illustration of PSGAN framework. MDNet distills makeup matrices from the reference image. AMM module applies the adapted makeup matrices to the output feature map of the third bottleneck of MANet to achieve makeup transfer. (B) Illustration of AMM module. Green blocks with 136 (68×2) channels indicate relative position features of the pixels, which are then concatenated with C -channel visual features. Thus, the attention map is computed for each pixel in the source image through the similarity of relative positions and visual appearances. The adapted makeup matrices γ' and β' are produced by the AMM module, which are then multiplied and added to feature maps of MANet element-wisely. The orange and the gray blocks in the figure indicate visual features with makeup and without makeup. (C) Attention maps for a specific red point in the source image. Note that we only calculate attentive values for pixels that belong to the same facial region. Thus, there are no response values on the lip and eye of the reference image.

can only transfer a fixed set of styles and cannot adapt to arbitrary new styles. Then, [13] proposed adaptive instance normalization (AdaIN) that aligns the mean and variance of the content features with those of the style features and achieved arbitrary style transfer. Here, we propose spatial-aware makeup transfer for each pixel rather than transferring a general style from the reference.

2.3. Attention Mechanism

Attention mechanism has been utilized in many areas [28, 22, 12, 23]. [26] proposed the attention mechanism in the natural language processing area by leveraging a self-attention module to compute the response at a position in a sequence (e.g., a sentence) by attending to all positions and taking their weighted average in an embedding space. [27] proposed the non-local network, which is to compute the response at a position as a weighted sum of the features at all positions. Inspired by these works, we explore the application of attention module by calculating the attention between two feature maps. Unlike the non-local network that only considers visual appearance similarities, our pro-

posed AMM module computes the weighted sum of another feature map by considering both visual appearances and locations.

3. PSGAN

3.1. Formulation

Let X and Y be the source image domain and the reference image domain. Also, we utilize $\{x^n\}_{n=1,\dots,N}$, $x^n \in X$ and $\{y^m\}_{m=1,\dots,M}$, $y^m \in Y$ to represent the examples of two domains respectively. Note that paired datasets are not required. That is, the source and reference images have different identities. We assume x is sampled from X according to the distribution $\mathcal{P}_X$ and y is sampled from Y according to the distribution $\mathcal{P}_Y$. Our proposed PSGAN learns a transfer function $G: \{x, y\} \rightarrow \tilde{x}$, where the transferred image $\tilde{x}$ has the makeup style of the reference image y and preserves the identity of the source image x .

3.2. Framework

Overall. The framework of PSGAN is shown in Figure 2 (A). Mathematically, it is formulated as $\tilde{x} = G(x, y)$. It

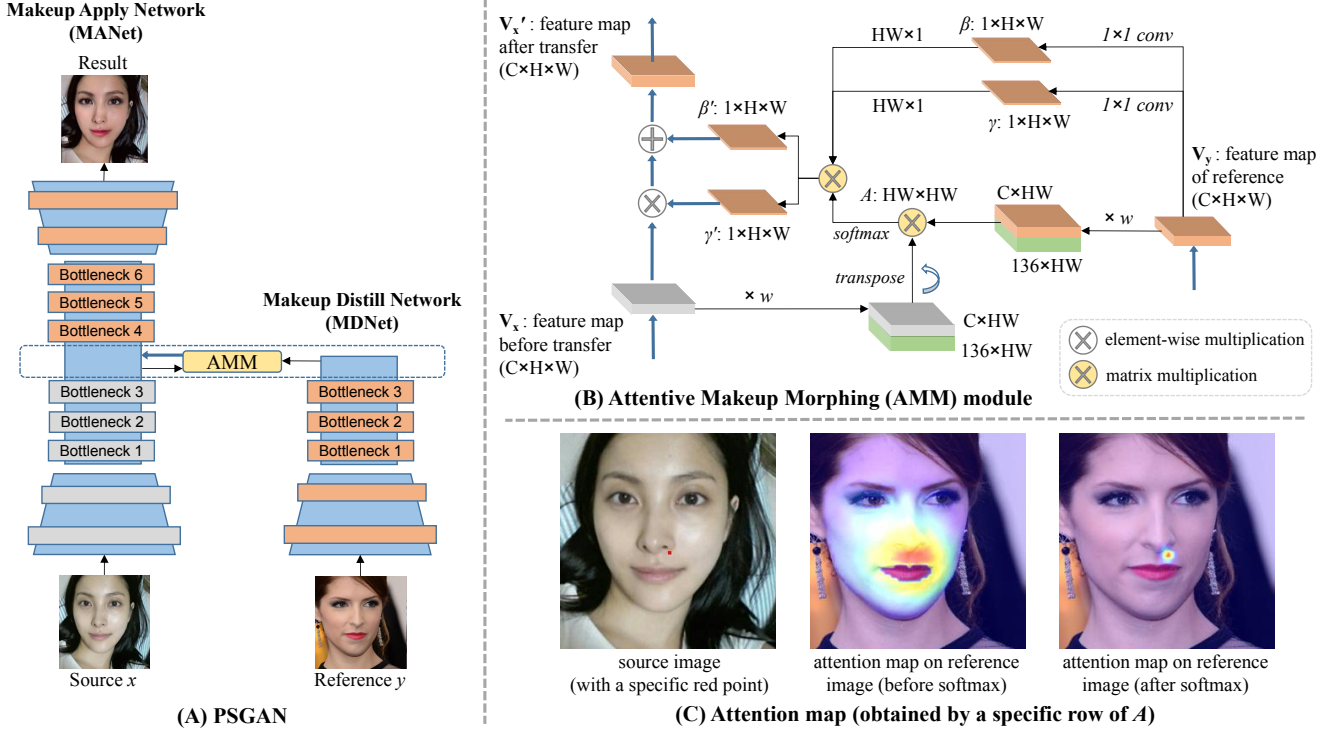


图2. (A) PSGAN框架示意图。MDNet从参考图像中提取化妆矩阵。AMM模块将适应后的化妆矩阵应用于MANet第三瓶颈层输出的特征图，从而实现妆容转移。(B) AMM模块示意图。绿色块表示具有136（68×2）通道的相对位置特征，这些特征随后与v1-通道视觉特征进行拼接。因此，通过相对位置和视觉外观的相似性计算源图像中每个像素的关注图。由AMM模块生成的适应后的化妆矩阵v2和v3将逐元素地乘加到MANet的特征图上。图中的橙色块和灰色块分别表示具有妆容和无妆容的视觉特征。(C) 源图像中特定红色点的关注图。请注意，我们仅计算属于同一面部区域的像素的关注值，因此参考图像上的唇部和眼部没有响应值。

只能传输一组固定的风格，无法适应任意新的风格。随后，[13]提出了自适应实例归一化（AdaIN）方法，该方法使内容特征的均值和方差与风格特征对齐，从而实现了任意风格的迁移。在这里，我们提出了一种针对每个像素的空间感知化妆传输方法，而不是从参考图像中传输通用风格。

2.3 注意力机制

注意力机制已在多个领域得到应用[28, 22, 12, 23]。文献[26]在自然语言处理领域提出了利用自注意模块计算序列（如句子）中某一位置响应的方法，该方法通过关注所有位置并在嵌入空间内取加权平均来实现。文献[27]则提出非局部网络，其计算方式为将所有位置的特征以加权的形式表示当前位置的响应。受这些工作的启发，我们探索了在两个特征图之间计算注意力的应用，并且与仅考虑视觉外观相似性的非局部网络不同，我们的方法——

提出的自适应内存管理模块通过考虑视觉外观和位置，计算另一个特征图的加权和。

3. PSGAN

3.1. 模型表述

令 (v_4) 和 (v_5) 分别表示源图像域和参考图像域。同时，我们使用 (v_6) 、 (v_7) 和 (v_8) 来分别表示两个域的示例。需要注意的是，并不要求成对的数据集，即源图像与参考图像是不同的身份。假设 (v_9) 是根据分布 (v_{11}) 从 (v_{10}) 中抽取的样本，而 (v_{12}) 是根据分布 (v_{14}) 从 (v_{13}) 中抽取的样本。我们提出的 PSGAN 学习了一个转换函数 $(v_{15} : v_{16})$ ，其中转换后的图像 (v_{17}) 具有参考图像 (v_{18}) 的妆容风格，同时保留了源图像 (v_{19}) 的身份特征。

3.2. 框架

整体而言，PSGAN的框架如图2 (A) 所示。从数学上讲，其形式化表示为 $\hat{x} = G(x, y)$ 。

can be divided into three parts. 1) *Makeup distill network*. MDNet extracts the makeup style from the reference image y and represents it as two makeup matrices γ and β , which have the same height and width as the feature map. 2) *Attentive makeup morphing module*. Since source images and reference images may have large discrepancies in expressions and poses, the extracted makeup matrices cannot be directly applied to the source image x . We then propose an AMM module to morph the two makeup matrices to two new matrices γ' and β' which are adaptive to the source image by considering the similarities between pixels of the source and reference. 3) *Makeup apply network*. The adaptive makeup matrices γ' and β' are applied to the bottleneck of the MANet to perform makeup transfer with pixel-level guidance by element-wise multiplication and addition.

Makeup distill network. The MDNet utilizes the encoder-bottleneck architecture used in [4] without the decoder part. It disentangles the makeup related features, e.g., lip gloss, eye shadows, from the intrinsic facial features, e.g., facial shape, the size of eyes. The makeup related features are represented as two makeup matrices γ and β , which are used to transfer the makeup by pixel-level operations. As shown in Figure 2 (B), the output feature map of MDNet $\mathbf{V}_y \in \mathbb{R}^{C \times H \times W}$ is fed into two 1×1 convolution layers to produce $\gamma \in \mathbb{R}^{1 \times H \times W}$ and $\beta \in \mathbb{R}^{1 \times H \times W}$, where C , H and W are the number of channels, height and width of the feature map.

Attentive makeup morphing module. Since the source and reference images may have different poses and expressions, the obtained spatial-aware γ and β cannot be applied directly to the source image. The proposed AMM module calculates an attentive matrix $A \in \mathbb{R}^{HW \times HW}$ to specify how a pixel in the source image x is morphed from the pixels in the reference image y , where $A_{i,j}$ indicates the attentive value between the i -th pixel x_i in image x and the j -th pixel y_j in image y .

Intuitively, makeup should be transferred between the pixels with similar relative positions on the face, and the attentive values between these pixels should be high. For example, the lip gloss region of the transferred result $\tilde{x}$ should be sampled from the corresponding lip gloss region of the reference image y . To describe the relative positions, we take the facial landmarks as anchor points. The relative position feature of pixel x_i is represented by $\mathbf{p}_i \in \mathbb{R}^{136}$, which is reflected in the differences of coordinates between pixel x_i and 68 facial landmarks, calculated by

$$\mathbf{p}_i = [f(x_i) - f(l_1), f(x_i) - f(l_2), \dots, f(x_i) - f(l_{68}), \\ g(x_i) - g(l_1), g(x_i) - g(l_2), \dots, g(x_i) - g(l_{68})], \quad (1)$$

where $f(\cdot)$ and $g(\cdot)$ indicate the coordinates on x and y axes, l_i indicates the i -th facial landmark obtained by the 2D facial landmark detector [29], which serves as the an-

chor point when calculating $\mathbf{p}_i$. In order to handle faces that occupy different sizes in images, we divide $\mathbf{p}$ by its two-norm (i.e., $\frac{\mathbf{p}}{\|\mathbf{p}\|}$) when calculating the attentive matrix.

Moreover, to avoid unreasonable sampling pixels with similar relative positions but different semantics, we also consider the visual similarities between pixels (e.g., x_i and y_j), which are denoted as the similarities between $\mathbf{v}_i$ and $\mathbf{v}_j$ that extracted from the third bottleneck of MANet and MDNet respectively. To make the relative position to be the primary concern, we multiply the visual features by a weight when calculating A . Then, the relative position features are resized and concatenated with the visual features along the channel dimension. As Figure 2 (B) shows, the attentive value $A_{i,j}$ is computed by considering the similarities of both visual appearances and relative positions via

$$A_{i,j} = \frac{\exp\left([w\mathbf{v}_i, \frac{\mathbf{p}_i}{\|\mathbf{p}_i\|}]^T [w\mathbf{v}_j, \frac{\mathbf{p}_j}{\|\mathbf{p}_j\|}]\right) \mathbb{I}(m_x^i = m_y^j)}{\sum_j \exp\left([w\mathbf{v}_i, \frac{\mathbf{p}_i}{\|\mathbf{p}_i\|}]^T [w\mathbf{v}_j, \frac{\mathbf{p}_j}{\|\mathbf{p}_j\|}]\right) \mathbb{I}(m_x^i = m_y^j)}, \quad (2)$$

where $[\cdot, \cdot]$ denotes the concatenation operation, $\mathbf{v} \in \mathbb{R}^C$ and $\mathbf{p} \in \mathbb{R}^{136}$ indicate the visual features and relative position features, w is the weight for visual features. $\mathbb{I}(\cdot)$ is an indicator function whose value is 1 if the inside formula is true, $m_x, m_y \in \{0, 1, \dots, N-1\}^{H \times W}$ are the face parsing map of source image x and reference image y , where N stands for the number of facial regions (N is 3 in our experiments including eyes, lip and skin), m_x^i and m_y^j indicate the facial regions that x_i and y_j belong to. Note that we only consider the pixels belonging to same facial region, i.e., $m_x^i = m_y^j$, by applying indicator function $\mathbb{I}(\cdot)$.

Given a specific point that marked in red in the lower-left corner of the nose in the source image, the middle image of Figure 2 (C) shows its attention map by reshaping a specific row of the attentive matrix $A_{i,:} \in \mathbb{R}^{1 \times HW}$ to $H \times W$. We can see that only the pixels around the left corner of the nose have large values. After applying softmax, attentive values become more gathered. This verifies that our proposed AMM module is able to locate semantically similar pixels to attend.

We multiply attentive matrix A by the γ and β , and get the morphed makeup matrices γ' and β' . More specifically, the matrices γ' and β' are computed by

$$\gamma'_i = \sum_j A_{i,j} \gamma_j; \\ \beta'_i = \sum_j A_{i,j} \beta_j, \quad (3)$$

where i and j are the pixel index of x and y . After that, the matrix $\gamma' \in \mathbb{R}^{1 \times H \times W}$ and $\beta' \in \mathbb{R}^{1 \times H \times W}$ are duplicated and expanded along the channel dimension to produce the makeup tensors $\Gamma' \in \mathbb{R}^{C \times H \times W}$ and $B' \in \mathbb{R}^{C \times H \times W}$, which will be the input of MANet.

可以分为三个部分。1) *Makeup distill network*. MDNet 从参考图像 y 中提取化妆风格，并将其表示为两个具有与特征图相同高度和宽度的化妆矩阵 γ 和 β 。2) *Attentive makeup morphing module*. 由于源图像和参考图像在表情和姿态上可能存在较大差异，直接将提取出的化妆矩阵应用于源图像 x 是不可行的。为此，我们提出了一种 AMM 模块，通过考虑源图与参考图像素之间的相似性，将两个化妆矩阵转换为适应源图像的新矩阵 γ' 和 β' 。3) *Makeup apply network*. 这些自适应化妆矩阵 γ' 和 β' 被应用于 MANet 的瓶颈层，通过逐元素乘法和加法操作，在像素级别指导下完成化妆迁移。

补妆蒸馏网络。MDNet 使用了文献 [4] 中的编码器瓶颈架构，但没有解码器部分。它将与化妆相关的特征（例如唇彩、眼影）从固有的面部特征（例如面部形状、眼睛大小）中分离出来。与化妆相关的特征表示为两个化妆矩阵 γ 和 β ，用于通过像素级操作传输妆容。如图 2 (B) 所示，MDNet 的输出特征图 $\mathbf{V}_y \in \mathbb{R}^{C \times H \times W}$ 被送入两个 $1 \times 1 \times 1$ 卷积层以生成 $\gamma \in \mathbb{R}^{1 \times H \times W}$ 和 $\beta \in \mathbb{R}^{1 \times H \times W}$ ，其中 C 、 H 和 W 分别表示特征图的通道数、高度和宽度。

注意力化妆变形模块。由于源图像和参考图像可能具有不同的姿态和表情，所获得的空间感知向量 γ 和 β 不能直接应用于源图像。提出的 AMM 模块计算一个注意力矩阵 $A \in \mathbb{R}^{HW \times HW}$ ，以指定源图像中的像素 x 如何从参考图像中的像素 y 变形，其中 $A_{i,j}$ 表示图像 x 中第 i 个像素 x_i 与图像 y 中第 j 个像素 y_j 之间的注意力值。

直观上，化妆效果应该在面部具有相似相对位置的像素之间进行转移，并且这些像素之间的注意力值应较高。例如，目标图像中的唇彩区域 $\tilde{x}$ 应当从参考图像中对应的唇彩区域 y 采样得到。为了描述相对位置，我们以面部特征点作为锚定点。像素 x_i 的相对位置特征由 $\mathbf{p}_i \in \mathbb{R}^{136}$ 表示，这反映了该像素与 68 个面部特征点之间的坐标差值，通过以下方式计算得出：

$$\mathbf{p}_i = [f(x_i) - f(l_1), f(x_i) - f(l_2), \dots, f(x_i) - f(l_{68}), g(x_i) - g(l_1), g(x_i) - g(l_2), \dots, g(x_i) - g(l_{68})], \quad (1)$$

其中， $f(\cdot)$ 和 $g(\cdot)$ 表示在 x 轴和 y 轴上的坐标， l_i 表示由二维面部关键点检测器 [29] 获取的第 i 个面部关键点，用作标注点。

在计算 $\mathbf{p}_i$ 时，为了处理占据不同大小的图像面片，我们在计算注意力矩阵时会将 $\mathbf{p}$ 除以其二范数（即 $\frac{\mathbf{p}}{\|\mathbf{p}\|}$ ）。

此外，为了避免采样像素具有相似相对位置但不同语义的问题，我们还考虑了像素之间的视觉相似性（例如， x_i 和 y_j ），这些相似性表示从 MANet 和 MDNet 的第三个瓶颈中提取的 $\mathbf{v}_i$ 与 $\mathbf{v}_j$ 之间的相似度。为了使相对位置成为主要关注点，在计算 A 时将视觉特征乘以一个权重。然后，将相对位置特征调整大小，并沿通道维度与视觉特征进行拼接。如图 2 (B) 所示，通过考虑视觉外观和相对位置的相似性来计算注意力值 $A_{i,j}$ 。

$$A_{i,j} = \frac{\exp\left([w\mathbf{v}_i, \frac{\mathbf{p}_i}{\|\mathbf{p}_i\|}]^T [w\mathbf{v}_j, \frac{\mathbf{p}_j}{\|\mathbf{p}_j\|}]\right) \mathbb{I}(m_x^i = m_y^j)}{\sum_j \exp\left([w\mathbf{v}_i, \frac{\mathbf{p}_i}{\|\mathbf{p}_i\|}]^T [w\mathbf{v}_j, \frac{\mathbf{p}_j}{\|\mathbf{p}_j\|}]\right) \mathbb{I}(m_x^i = m_y^j)}, \quad (2)$$

其中 $[\cdot, \cdot]$ 表示连接操作， $\mathbf{v} \in \mathbb{R}^C$ 和 $\mathbf{p} \in \mathbb{R}^{136}$ 分别表示视觉特征和相对位置特征， w 是视觉特征的权重。 $\mathbb{I}(\cdot)$ 是一个指示函数，当内部公式为真时其值为 1。 $m_x, m_y \in \{0, 1, \dots, N-1\}^{H \times W}$ 分别是源图像 x 和参考图像 y 的面部解析图，其中 N 表示面部区域的数量（在我们的实验中包括眼睛、嘴唇和皮肤，因此 N 等于 3）， m_x^i 和 m_y^j 分别表示 x_i 和 y_j 所属的面部区域。需要注意的是，我们仅考虑属于同一面部区域的像素，即通过应用指示函数 $\mathbb{I}(\cdot)$ 来实现这一目的。

给定源图像中鼻子左下角标记为红色的特定点，在图 2(C) 中的中间图像展示了其注意力图，通过将关注矩阵的特定行 $A_{i,:} \in \mathbb{R}^{1 \times HW}$ 重塑为 $H \times W$ 。可以看到只有鼻子左角周围的像素值较大。经过 softmax 处理后，关注值更加集中，这验证了我们提出的 AMM 模块能够定位语义相似的像素进行关注。

我们将注意力矩阵 A 与 γ 和 β 相乘，得到变形后的妆容矩阵 γ' 和 β' 。更具体地说，矩阵 γ' 和 β' 的计算方式为

$$\begin{aligned} \gamma'_i &= \sum_j A_{i,j} \gamma_j; \\ \beta'_i &= \sum_j A_{i,j} \beta_j, \end{aligned} \quad (3)$$

其中 i 和 j 分别是 x 和 y 的像素索引。之后，矩阵 $\gamma' \in \mathbb{R}^{1 \times H \times W}$ 和 $\beta' \in \mathbb{R}^{1 \times H \times W}$ 沿通道维度进行复制和扩展，生成化妆张量 $\Gamma' \in \mathbb{R}^{C \times H \times W}$ 和 $B' \in \mathbb{R}^{C \times H \times W}$ ，这些将作为 MANet 的输入。

Makeup apply network. MANet utilizes a similar encoder-bottleneck-decoder architecture as [4]. As shown in Figure 2 (A), the encoder part of MANet shares the same architecture with MDNet, but they do not share parameters. In the encoder part, we use instance normalizations that have no affine parameters that make the feature map to be a normal distribution. In the bottleneck part, the morphed makeup tensors Γ' and B' obtained by the AMM module are applied to the source image feature map $\mathbf{V}_x \in \mathbb{R}^{C \times H \times W}$. The activation values of the transferred feature map $\mathbf{V}_x'$ are calculated by

$$\mathbf{V}_x' = \Gamma' \mathbf{V}_x + B'. \quad (4)$$

Eq. (4) gives the function of makeup transfer. The updated feature map $\mathbf{V}_x'$ is then fed to the subsequent decoder part of MANet to produce the transferred result.

3.3. Objective Function

Adversarial loss. We utilize two discriminators D_X and D_Y for the source image domain X and the reference image domain Y , which try to discriminate between generated images and real images and thus help the generators synthesize realistic outputs. Therefore, the adversarial loss L_D^{adv} , L_G^{adv} for discriminator and generator are computed by

$$\begin{aligned} L_D^{adv} &= -\mathbb{E}_{x \sim \mathcal{P}_X} [\log D_X(x)] - \mathbb{E}_{y \sim \mathcal{P}_Y} [\log D_Y(y)] \\ &\quad - \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (1 - D_X(G(y, x)))] \\ &\quad - \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (1 - D_Y(G(x, y)))] \\ L_G^{adv} &= -\mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (D_X(G(y, x)))] \\ &\quad - \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (D_Y(G(x, y)))] \end{aligned} \quad (5)$$

Cycle consistency loss. Due to the lack of triplets data (source image, reference image, and transferred image), we train the network in an unsupervised way. Here, we introduce the cycle consistency loss proposed by [31]. We use the L1 loss to constrain the reconstructed images and define the cycle consistency loss L_G^{cyc} as

$$\begin{aligned} L_G^{cyc} &= \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(G(x, y), x) - x\|_1] \\ &\quad + \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(G(y, x), y) - y\|_1]. \end{aligned} \quad (6)$$

Perceptual loss. When transferring the makeup style, the transferred image is required to preserve personal identity. Instead of directly measuring differences at pixel-level, we utilize a VGG-16 model pre-trained on ImageNet to compare the activations of source images and generated images in the hidden layer. Let $F_l(\cdot)$ denote the output of the l -th layer of VGG-16 model. We introduce the perceptual loss L_G^{per} to measure their differences using L2 loss:

$$\begin{aligned} L_G^{per} &= \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|F_l(G(x, y)) - F_l(x)\|_2] \\ &\quad + \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|F_l(G(y, x)) - F_l(y)\|_2]. \end{aligned} \quad (7)$$

Makeup loss. To provide coarse guidance for makeup transfer, we utilize the makeup loss proposed by [17]. Specifically, we perform histogram matching on the same facial regions of x and y separately and then recombine the results, denoted as $HM(x, y)$. As a kind of pseudo ground truth, $HM(x, y)$ preserves the identity of x and has a similar color distribution with y . Then we calculate the makeup loss L_G^{make} as coarse guidance by

$$\begin{aligned} L_G^{make} &= \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(x, y) - HW(x, y)\|_2] \\ &\quad + \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(y, x) - HW(y, x)\|_2]. \end{aligned} \quad (8)$$

Total loss. The loss L_D and L_G for discriminator and generator of our approach can be expressed as

$$\begin{aligned} L_D &= \lambda_{adv} L_D^{adv} \\ L_G &= \lambda_{adv} L_G^{adv} + \lambda_{cyc} L_G^{cyc} + \lambda_{per} L_G^{per} + \lambda_{make} L_G^{make}, \end{aligned} \quad (9)$$

where λ_{adv} , λ_{cyc} , λ_{per} , λ_{make} are the weights to balance the multiple objectives.

4. Experiments

4.1. Data Collection

Since the existing makeup datasets only consist of frontal facial images with neutral expressions, we collect a new Makeup-Wild dataset that contains facial images with various poses and expressions as well as complex backgrounds to test methods in the real-world environment. We collect data from the Internet and then manually remove images with frontal face or neutral expression. After that, we crop and resize the images to be 256×256 resolution without alignment. Finally, 403 with-makeup images and 369 non-makeup images are collected to form the Makeup-Wild dataset.

4.2. Experimental Setting and Details

We train our network using the training part of the MT (Makeup Transfer) dataset [17, 3] and test it on the testing part of MT dataset and the Makeup-Wild dataset. MT dataset contains 1,115 non-makeup images and 2,719 with-makeup images which are mostly well-aligned, with the resolution of 361×361 and the corresponding face parsing results. We follow the splitting strategy of [17] to form the train/test set and conduct frontal face experiments in the test set of MT dataset since the examples in the test set are well-aligned frontal facial images. To further prove the effectiveness of PSGAN for handling pose and expression differences, we use the Makeup-Wild dataset as an extra test set. Note that we only train our network using the training part of the MT dataset for a fair comparison.

For all experiments, we resize the images to 256×256 , and utilize the *relu_4_1* feature layer of VGG-16 for calcu-

Makeup应用网络。MANet采用与[4]类似的编码器-瓶颈-解码器架构。如图2 (A) 所示, MANet的编码器部分与MDNet具有相同的架构, 但它们不共享参数。在编码器部分, 我们使用实例归一化操作, 该操作没有仿射参数, 使得特征图服从正态分布。在瓶颈部分, 通过AMM模块获得的变形妆容张量 Γ' 和 B' 被应用于源图像特征图 $\mathbf{V}_x \in \mathbb{R}^{C \times H \times W}$ 。传输后的特征图 $\mathbf{V}_x'$ 的激活值由以下方式计算:

$$\mathbf{V}_x' = \Gamma' \mathbf{V}_x + B'. \quad (4)$$

公式(4)给出了 Makeup Transfer 的函数。更新后的特征图 $\mathbf{V}_x'$ 然后被输入到 MANet 的后续解码部分, 以生成转换结果。

3.3 目标函数

对抗损失。我们使用两个判别器 D_X 和 D_Y 分别针对源图像域 X 和参考图像域 Y , 它们试图区分生成的图像与真实的图像, 从而帮助生成器合成逼真的输出结果。因此, 判别器和生成器的对抗损失 L_D^{adv} 、 L_G^{adv} 通过以下方式计算:

$$\begin{aligned} L_D^{adv} &= -\mathbb{E}_{x \sim \mathcal{P}_X} [\log D_X(x)] - \mathbb{E}_{y \sim \mathcal{P}_Y} [\log D_Y(y)] \\ &\quad - \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (1 - D_X(G(y, x)))] \\ &\quad - \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (1 - D_Y(G(x, y)))] \\ L_G^{adv} &= -\mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (D_X(G(y, x)))] \\ &\quad - \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\log (D_Y(G(x, y)))] \end{aligned} \quad (5)$$

周期一致性损失。由于缺乏三元组数据(源图像、参考图像和转移图像), 我们采用无监督方式训练网络。在这里, 我们引入了由[31]提出的周期一致性损失。我们使用L1损失来约束重构的图像, 并定义周期一致性损失 L_G^{cyc} 为

$$\begin{aligned} L_G^{cyc} &= \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(G(x, y), x) - x\|_1] \\ &\quad + \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(G(y, x), y) - y\|_1]. \end{aligned} \quad (6)$$

感知损失。在转移化妆风格时, 要求转换后的图像保留个人身份特征。我们不直接测量像素级别的差异, 而是利用在ImageNet上预训练的VGG-16模型, 在隐藏层比较源图像和生成图像的激活值。设 $F_l(\cdot)$ 表示VGG-16模型第 l 层的输出。我们引入感知损失 L_G^{per} , 使用L2损失来衡量它们之间的差异:

$$\begin{aligned} L_G^{per} &= \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|F_l(G(x, y)) - F_l(x)\|_2] \\ &\quad + \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|F_l(G(y, x)) - F_l(y)\|_2]. \end{aligned} \quad (7)$$

补妆损失。为了提供粗略的补妆转移指导, 我们采用了[17]提出的补妆损失。具体来说, 我们在 x 和 y 的相同面部区域分别进行直方图匹配, 然后重新组合结果, 记为 $HM(x, y)$ 。作为一种伪真实标签, $HM(x, y)$ 保留了 x 的身份特征, 并且与 y 具有相似的颜色分布。随后, 我们通过计算补妆损失 L_G^{make} 作为粗略指导。

$$\begin{aligned} L_G^{make} &= \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(x, y) - HW(x, y)\|_2] \\ &\quad + \mathbb{E}_{x \sim \mathcal{P}_X, y \sim \mathcal{P}_Y} [\|G(y, x) - HW(y, x)\|_2]. \end{aligned} \quad (8)$$

总损失。我们方法中的判别器和生成器的损失分别表示为 L_D 和 L_G 。

$$\begin{aligned} L_D &= \lambda_{adv} L_D^{adv} \\ L_G &= \lambda_{adv} L_G^{adv} + \lambda_{cyc} L_G^{cyc} + \lambda_{per} L_G^{per} + \lambda_{make} L_G^{make}, \end{aligned} \quad (9)$$

其中 λ_{adv} 、 λ_{cyc} 、 λ_{per} 和 λ_{make} 是用于平衡多个目标的权重。

4. 实验

4.1. 数据收集

由于现有的化妆数据集仅包含正面且表情自然的脸部图像, 我们收集了一个新的 Makeup-Wild 数据集, 其中包含了各种姿势、表情以及复杂背景的面部图像, 以在真实环境中测试方法。我们从互联网上收集数据, 并手动移除正面或表情自然的照片。之后, 我们将图像裁剪并调整为 256×256 分辨率, 无需对齐处理。最终, 该 Makeup-Wild 数据集包含了 403 张化妆后的图像和 369 张未化妆的图像。

4.2. 实验设置与细节

我们使用MT(妆容转换)数据集的训练部分[17,3]来训练网络, 并在MT数据集和Makeup-Wild数据集的测试部分上进行测试。MT数据集包含1,张非化妆图像和2,张化妆图像, 这些图像大多对齐良好, 分辨率为 361×361 , 并附有相应的人脸解析结果。我们遵循[17]中的划分策略来形成训练集和测试集, 并在MT数据集的测试集中进行正面人脸实验, 因为该测试集中的示例是对齐良好的正面面部图像。为了进一步证明PSGAN处理姿态和表情差异的有效性, 我们将Makeup-Wild数据集用作额外的测试集。请注意, 我们仅使用MT数据集的训练部分来公平地比较我们的网络性能。

对于所有实验, 我们将图像调整为256×256大小, 并使用VGG-16的第4个1×1特征层进行计算。



Figure 3. Without AMM module, the makeup transfer results (the 3rd column) are bad due to pose and expression differences between source and reference images.

lating perceptual loss. The weights of different loss functions are set as $\lambda_{adv} = 1$, $\lambda_{cyc} = 10$, $\lambda_{per} = 0.005$, $\lambda_{make} = 1$, and the weight for visual feature in AMM is set to 0.01. We train the model for 50 epochs optimized by Adam [15] with learning rate of 0.0002 and batch size of 1.

4.3. Ablation Studies

Attentive makeup morphing module. In PSGAN, AMM module morphs the distilled makeup matrices γ and β to γ' , β' . It alleviates the pose and expression differences between source and reference images. The effectiveness of the AMM module is shown in Figure 3. In the first row, the pose of source and reference images are very different. The bangs of the reference image are transferred to the skin of the source image without AMM. By applying AMM, the pose misalignment is well solved. A similar observation can be found in the second row: the expressions of source and reference images are smiling and neutral respectively, while the lip gloss is applied to the teeth region without the AMM module shown in the third column. After integrating AMM, lip gloss is applied to the lip region, bypassing the teeth area. The experiments demonstrate that the AMM module can specify how a pixel in the source image is morphed from pixels of the reference instead of mapping the makeup from the same location directly.

The weight of visual feature in calculating A . In the AMM module, we calculate the attentive matrix A by considering both the visual features $\mathbf{v}$ and relative positions $\mathbf{p}$ using Eq. (2). Figure 4 demonstrates that if only relative positions are considered by setting the weight to zero, the attentive maps in the second column are similar to a 2D Gaussian distribution. In the first column of Figure 4, the red point on the skin of the source may wrongly receive makeup from the nostrils area in the reference image (1st row). The attention map also crosses the face boundary and covers the earrings (2nd row) which is unreasonable. Besides, larger weights will lead to scattered and unreasonable attention maps, as shown in the last column. After con-

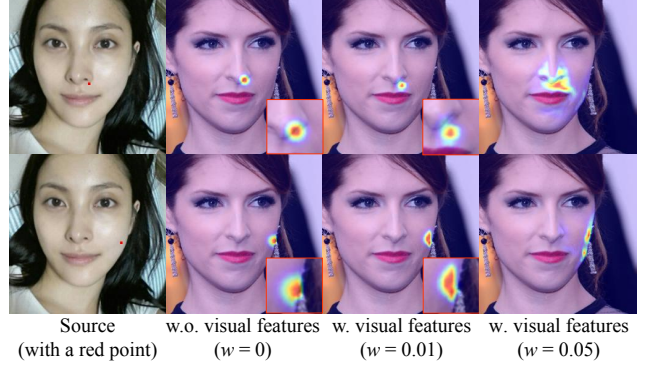


Figure 4. Given a red point on the skin, the corresponding attention maps with different weights on visual features are shown. Without using visual features, attention maps fail to avoid nostrils (1st row, 2nd column) and wrongly crosses the facial boundary (2nd row, 2nd column). While a larger weight leads to scattered and unreasonable attention maps.



Figure 5. Given the source image (2nd column), the transferred images (3rd column) are generated by transferring the lipstick from reference 1 and other makeup from reference 2.

sidering the appearance feature appropriately by setting the weight to 0.01, the attention maps focus more on the skin and also bypass the nostrils as well as background.

4.4. Partial and Interpolated Makeup Transfer

Since the makeup matrices γ and β are spatial-aware, the partial and interpolated transfer can be realized during testing. To achieve partial makeup generation, we compute the new makeup matrices by weighting the matrices using the face parsing results. Let x , y_1 , and y_2 denote a source image and two reference images. We can obtain Γ'_x , B'_x and Γ'_{y_1} , B'_{y_1} as well as Γ'_{y_2} , B'_{y_2} by feeding the images to MDNet. In addition, we can obtain the face parsing mask m_x of x through the existing deep learning method [30]. Suppose we want to mix the lipstick from y_1 and other makeup from y_2 , we can first obtain the binary mask of the lip, denoted as $m_x^l \in \{0, 1\}^{H \times W}$. Then, PSGAN can realize partial makeup transfer by assigning different makeup parameters on different pixels. By modifying Eq. (4), the partial transferred feature map $\mathbf{V}_x'$ can be calculated by

$$\mathbf{V}_x' = (m_x^l \Gamma'_{y_1} + (1 - m_x^l) \Gamma'_{y_2}) \mathbf{V}_x + (m_x^l B'_{y_1} + (1 - m_x^l) B'_{y_2}). \quad (10)$$

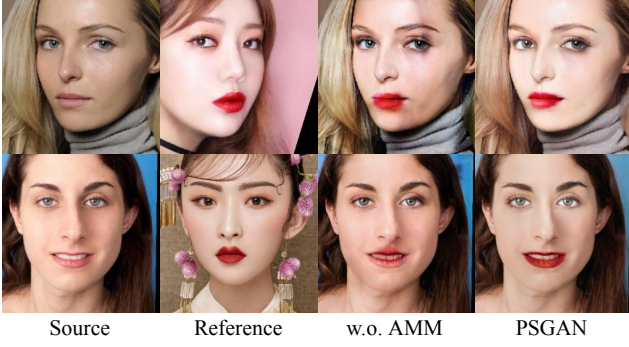


图3. 未使用AMM模块时，由于源图像和参考图像在姿态和表情上的差异，补全传输结果（第3列）较差。

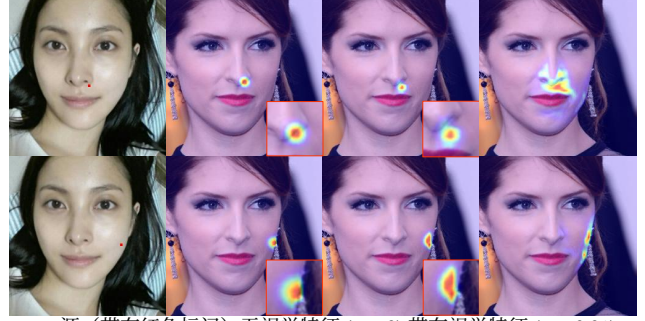
为了降低感知损失，不同损失函数的权重设置为

$\lambda_{adv} = 1, \lambda_{cyc} = 10, \lambda_{per} = 0.0005, \lambda_{make} = 1$ ，而AMM中视觉特征的权重设为0.01。模型通过Adam [15] 优化器训练50个周期，学习率为0.0002，批处理大小为1。

4.3. 消融实验

注意力化妆变形模块。在PSGAN中，AMM模块将提取的妆容矩阵 γ 和 β 转换为 γ' 和 β' 。它缓解了源图像与参考图像之间的姿态和表情差异。图3展示了AMM模块的有效性。第一行显示源图像和参考图像的姿态相差很大。在没有使用AMM的情况下，参考图像的刘海被转移到了源图像的脸部皮肤上。通过应用AMM，姿态错配问题得到了很好的解决。第二行也有类似观察：源图像的表情是微笑的，而参考图像是中性的表情，在未使用AMM模块的情况下（第三列），唇膏涂抹到了牙齿区域。在集成AMM后，唇膏被正确地涂抹到嘴唇区域，并避开牙齿部分。实验表明，AMM模块可以指定源图像中的一个像素如何从参考图像中的多个像素变形而来，而不是直接映射同一位置的妆容。

视觉特征在计算 A 中的权重。在AMM模块中，我们通过考虑视觉特征 v 和相对位置 p 来计算注意力矩阵 A （见公式(2)）。图4显示，如果仅考虑相对位置并将权重设为零，则第二列的注意力图类似于二维高斯分布。在图4的第一列中，源图像皮肤上的红色点可能会错误地从参考图像鼻孔区域接收妆容（第一行）。注意力图还跨越面部边界并覆盖耳环（第二行），这是不合理的。此外，更大的权重会导致分散且不合理的结果，如最后一列所示。继续



源（带有红色标记）无视觉特征 ($w = 0$) 带有视觉特征 ($w = 0.01$) 带有视觉特征 ($w = 0.05$)

图4. 给定皮肤上的一个红点，展示了不同权重下视觉特征对应的注意力图。不使用视觉特征时，注意力图无法避开鼻孔（第1行，第2列）并且错误地跨越了面部边界（第2行，第2列）。而更大的权重会导致分散且不合理的结果。



图5. 给定源图像（第2列），转移后的图像（第3列）是通过将口红从参考1进行转移，其他化妆品从参考2进行转移而生成的。

通过将权重设置为0.01，可以适当考虑外观特征，从而使注意力图更侧重于皮肤，并且避开鼻孔及背景。

4.4. 部分及插值妆容转移

由于Makeup矩阵 γ 和 β 具有空间感知能力，因此可以在测试过程中实现部分和插值的转移。为了实现部分Makeup的生成，我们通过使用面部解析结果对矩阵进行加权来计算新的Makeup矩阵。令 x 、 y_1 和 y_2 分别表示源图像和两个参考图像。将这些图像输入MDNet后，我们可以获得 Γ'_x 、 B'_x 以及 Γ'_{y_2} 、 B'_{y_2} 。此外，我们还可以通过现有的深度学习方法 [30] 获得 x 的面部解析掩码 m_x 。假设我们希望将来自 y_1 的唇膏和其他来自 y_2 的Makeup混合在一起，我们可以首先获得嘴唇的二值掩码，表示为 $m_x^l \in \{0, 1\}^{H \times W}$ 。然后PSGAN可以通过在不同像素上分配不同的Makeup参数来实现部分Makeup转移。通过对公式(4)进行修改，可以计算出部分转移的特征图 V_x' 。

$$V_x' = (m_x^l \Gamma'_{y_1} + (1 - m_x^l) \Gamma'_{y_2}) V_x + (m_x^l B'_{y_1} + (1 - m_x^l) B'_{y_2}). \quad (10)$$

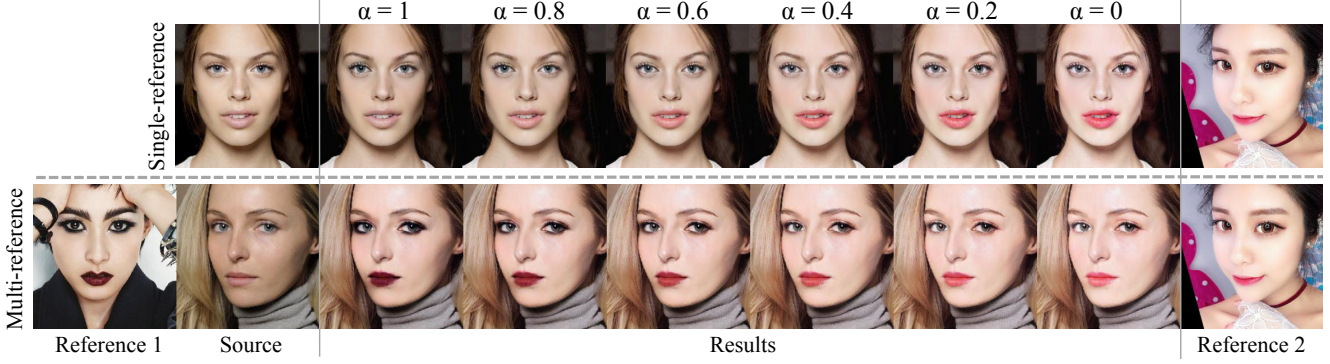


Figure 6. Results of interpolated makeup styles. If only one reference is used, adjusting the shade of makeup can be realized (1st row). If two references are used (1st column and last column), the makeup of the transferred images is gradually changing from reference 1 towards reference 2 from left to right (2nd rows).

Figure 5 shows the results by mixing the makeup styles from two references partially. The results on the third column recombine the makeup of lip from reference 1 and other part of makeup from reference 2, which are natural and realistic. Also, only transferring the lipstick from reference 1 and remain other parts unchanged can be achieved by assigning $x = y_2$. The new feature of partial makeup makes PSGAN realize the flexible partial makeup transfer.

Moreover, we can interpolate the makeup with two reference images by a coefficient $\alpha \in [0, 1]$. We first get the makeup tensors of two references y_1 and y_2 , and then compute the new parameters by weighting them with the coefficient α . The resulted feature map $\mathbf{V}_{\mathbf{x}'}$ is calculated by

$$\mathbf{V}_{\mathbf{x}'} = (\alpha\Gamma'_{y_1} + (1 - \alpha)\Gamma'_{y_2})\mathbf{V}_{\mathbf{x}} + (\alpha B'_{y_1} + (1 - \alpha)B'_{y_2}). \quad (11)$$

Figure 6 shows the interpolated makeup transfer results with one and two reference images. By feeding the new makeup tensors into MANet, we yield a smooth transition between two reference makeup styles. Similarly, we can adjust the shade of transfer using only one reference image by assigning $x = y_1$. The generated results demonstrate that our PSGAN can not only control the shade of makeup transfer but also generate a new style of makeup by mixing the makeup tensors of two makeup styles.

We can also perform partial and interpolated transfer simultaneously by leveraging both the face parsing maps and coefficient thanks to the design of spatial-aware makeup matrices. The above experiments have demonstrated that PSGAN broadens the application range of makeup transfer significantly.

4.5. Comparison

We conduct comparison with general image-to-image translation methods DIA [18] and CycleGAN [31] as well as state-of-the-art makeup transfer methods BeautyGAN (BGAN) [17], PairedCycleGAN (PGAN) [2], BeautyGlow (BGlow) [3] and LADN [10]. Current makeup transfer

Method	Functions		
	Shade	Part	Robust
BGAN [17]			
PGAN [2]			
BGlow [3]	✓		
LADN [10]	✓		
PSGAN	✓	✓	✓

Table 1. Analyses of existing methods. “Shade”, “Part” and “Robust” indicate shade-controllable, partial and pose/expression robust transfer respectively.

Test set	PSGAN	BGAN	DIA	CGAN	LADN
MT	61.5	32.5	3.25	2.5	0.25
M-Wild	83.5	13.5	1.75	1.25	0.0

Table 2. Ratio selected as best (%).

methods leverage face parsing maps [2, 3, 17] and facial landmarks [10] for training and realize different functions as shown in Table 1.

Quantitative Comparison. We conduct a user study for quantitative evaluation on Amazon Mechanical Turk (AMT) that use BGAN, CGAN, DIA, and LADN as baselines. For a fair comparison, we only compare with methods whose code and pre-train model are released since we cannot guarantee a perfect re-implementation. We randomly select 20 source images and 20 reference images from both the MT test set and Makeup-Wild (M-Wild) dataset. After using the above methods to perform makeup transfer between these images, we obtain 800 images for each method. Then, 5 different workers are asked to choose the best images generated by five methods through considering image realism and the similarity with reference makeup styles. The generated images are shown in random order for a fair comparison. Table 2 shows the human evaluation results.

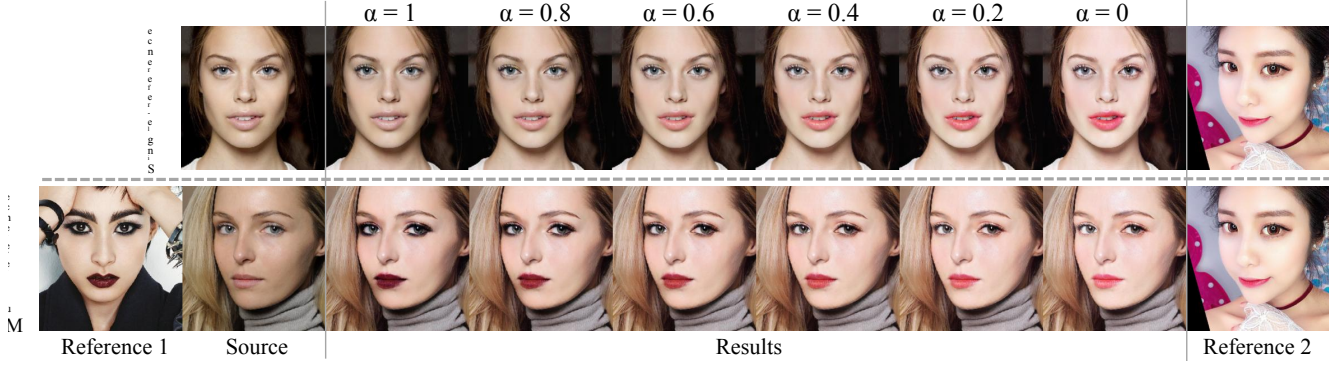


图6. 插值化妆风格的结果。若仅使用一个参考图像，则可以调整妆容的深浅（第一行）。若使用两个参考图像（第一列和最后一列），则从左到右，转移图像中的妆容会逐渐从参考1过渡至参考2（第二行）。

图5展示了将两个参考样例的化妆风格部分混合后的结果。第三列的结果重新组合了来自参考1的唇部妆容和来自参考2的其他部位妆容，效果自然且逼真。此外，仅转移参考1的口红并将其他部分保持不变也可以通过分配 $x = y_2$ 实现。这种局部化妆的新特性使得PSGAN能够实现灵活的部分化妆迁移。

此外，我们可以通过一个系数 $\alpha \in [0,1]$ 使用两张参考图像插值出妆容。首先获取两张参考图像的妆容张量 y_1 和 y_2 ，然后通过该系数 α 权重计算新的参数。最终得到的新特征图 $\mathbf{V}_x'$ 由以下方式计算得出。

$$\mathbf{V}_x' = (\alpha \Gamma'_{y_1} + (1 - \alpha) \Gamma'_{y_2}) \mathbf{V}_x + (\alpha B'_{y_1} + (1 - \alpha) B'_{y_2}). \quad (11)$$

图6展示了使用一张和两张参考图像进行插值化妆转移的结果。通过将新的化妆张量输入到MANet中，我们能够实现两种参考化妆风格之间的平滑过渡。同样地，仅使用一张参考图像时，可以通过分配 $x = y_1$ 来调整转移的色调。生成的结果表明，我们的PS-GAN不仅可以控制化妆转移的色调，还可以通过混合两种化妆风格的化妆张量来生成新的化妆风格。

通过利用面部解析图和系数，我们可以借助空间感知妆容矩阵的设计同时进行部分转移和插值转移。上述实验已经证明，PSGAN显著扩大了妆容转移的应用范围。

4.5. 对比分析

我们与通用图像到图像翻译方法DIA [18]和CycleGAN [31]以及当前最先进的化妆转移方法BeautyGAN (BGAN) [17]、PairedCycleGAN (PGAN) [2]、BeautyGlow (BGlow) [3]和LADN [10]进行了比较。目前的化妆转移技术

Method	Functions		
	Shade	Part	Robust
BGAN [17]			
PGAN [2]			
BGlow [3]	✓		
LADN [10]	✓		
PSGAN	✓	✓	✓

表1. 现有方法的分析。“Shade”，“Part”和“Robust”分别表示可调节阴影、部分转移和姿态/表情鲁棒性转移。

Test set	PSGAN	BGAN	DIA	CGAN	LADN
MT	61.5	32.5	3.25	2.5	0.25
M-Wild	83.5	13.5	1.75	1.25	0.0

表2. 选为最佳的比例 (%)

方法利用面部解析图[2, 3, 17]和面部特征点[10]进行训练，实现不同的功能，如表1所示。

定量比较。我们通过在亚马逊机械土耳其人 (AMT) 上进行用户研究，使用BGAN、CGAN、DIA和LADN作为基准方法来进行定量评估。为了公平对比，我们仅与那些代码和预训练模型已公开的方法进行比较，因为我们无法保证完美重现其他方法的实现。从MT测试集和化妆野外 (M-Wild) 数据集中随机选取20张源图像和20张参考图像。使用上述方法在这两组图像之间执行化妆转换后，我们为每种方法获得了800张生成的图像。然后，要求5名不同的工作者在考虑图像真实感和与参考妆容风格相似度的情况下，从五种方法中选择最佳生成图像。为了公平比较，这些生成的图像以随机顺序展示。表2显示了人类评估的结果。

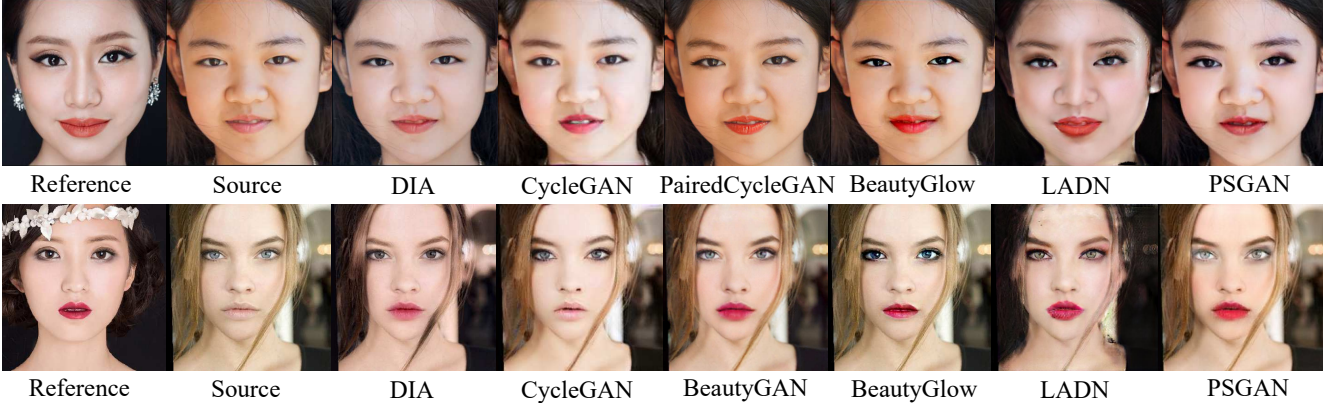


Figure 7. Qualitative comparison. PSGAN is able to generate realistic images with the same makeup styles as the reference.

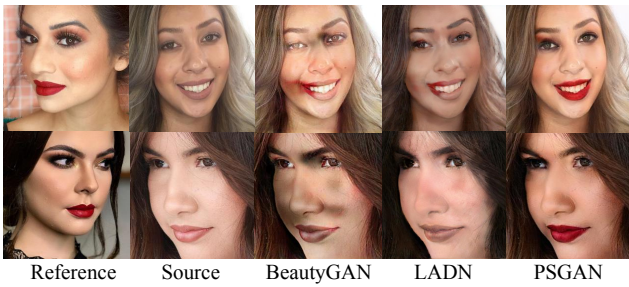


Figure 8. Qualitative comparison on M-Wild test set.

Our PSGAN outperforms other methods by a large margin, especially on the M-Wild test set.

Qualitative Comparison. Figure 7 shows the qualitative comparison of PSGAN with other state-of-the-art methods on frontal faces in neutral expressions. Since the code of BeautyGlow and PairedCycleGAN is not released, we follow the strategy of BeautyGlow which cropped the results from corresponding papers. The result produced by DIA has an unnatural color on hair and background since it performs transfer in the whole image. Comparatively, the result of CycleGAN is more realistic than that of DIA, but CycleGAN can only synthesize general makeup which is not similar to the reference. The current makeup transfer methods outperform the previous methods. However, BeautyGlow fails to preserve the color of pupils and does not have the same foundation makeup as reference. We also use the pre-trained model released by the author of LADN, which produces blurry transfer results and unnatural background. Compared to the baselines, our method is able to generate vivid images with the same makeup styles as reference.

We also conduct a comparison on the M-Wild test set with the state-of-the-art method (BeautyGAN and LADN) that provide code and pre-trained model, as shown in Figure 8. Since the current methods lack an explicit mechanism to guide the direction of make transfer at the pixel-level and also overfit on frontal images, the makeup is applied in the



Figure 9. Video makeup transfer results of PSGAN.

wrong region of the face when dealing with images with different poses and expressions. For example, the lip gloss is transferred to the skin on the first row of Figure 8. In the second row, other methods fail to perform transfer on faces with different sizes. However, the proposed AMM module can accurately assign the makeup for every pixel through calculating the similarities, which makes our results look better.

4.6. Video Makeup Transfer

To transfer makeup for a person in the video is a challenging and meaningful task, which has wide prospects in the applications. However, the pose and expression of a face in the video are continuously changing which brings extra difficulties. To examine the effectiveness of our method, we simply perform makeup transfer on every frame of the video, as shown in Figure 9. By incorporating the design of PSGAN, we receive nice and stable transferred results.

5. Conclusion

In order to bring makeup transfer to real-world applications, we propose the Pose and expression robust Spatial-Aware GAN (PSGAN) that first distills the makeup style into two makeup matrices from the reference and then

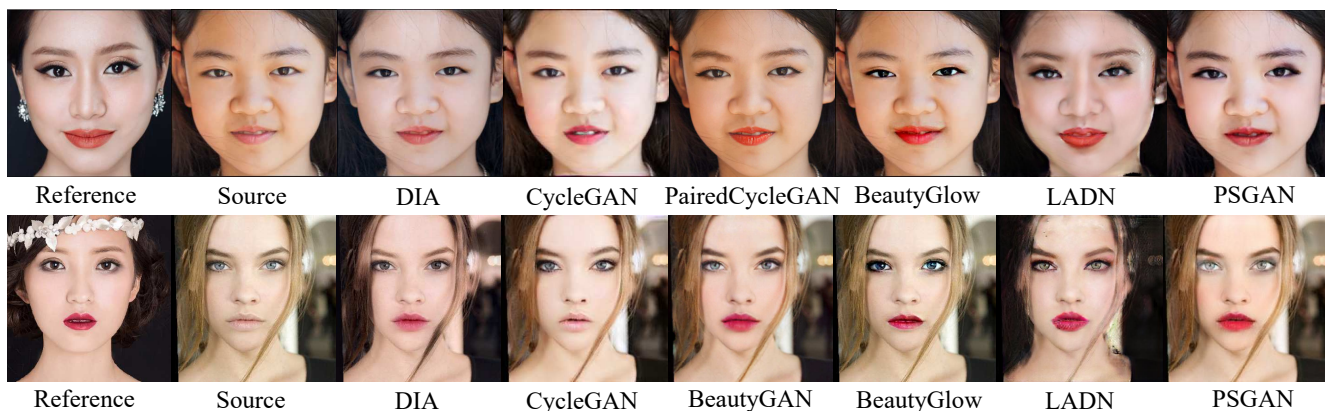


图7. 定性比较。PSGAN能够生成与参考图像具有相同妆容风格的逼真图片。

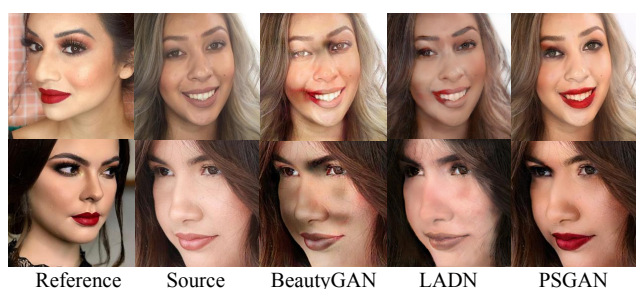


图8. 在M-Wild测试集上的定性比较。

我们的PSGAN在各项方法中表现优异，特别是在M-Wild测试集上。

定性比较。图7展示了PSGAN与其他最新方法在中性表情正面人脸上的定性对比。由于BeautyGlow和PairedCycleGAN的代码未公开，我们采用了与BeautyGlow相同的方法，从相关论文中裁剪结果进行展示。DIA生成的结果在头发和背景上颜色不自然，因为它在整个图像上执行了迁移操作。相比之下，CycleGAN的结果比DIA更逼真，但CycleGAN只能合成一般化的妆容，与参考图片并不相似。当前的化妆迁移方法优于以往的方法。然而，BeautyGlow未能保留瞳孔的颜色，并且没有与参考图相同的底妆。我们还使用了LADN作者发布的预训练模型，该模型生成的结果模糊不清并且背景不自然。相比基准方法，我们的方法能够生成逼真的图像，具有与参考图片一致的化妆风格。

我们也对 M-Wild 测试集与当前最先进的方法（BeautyGAN 和 LADN）进行了比较，如图8所示。由于现有方法缺乏明确的机制来指导像素级别的化妆转移方向，并且在正面图像上过度拟合，因此化妆是在此基础上应用的。

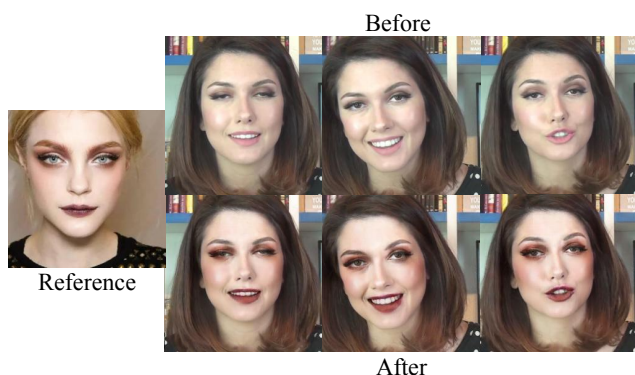


图9. PSGAN的视频妆容迁移结果。

在处理不同姿态和表情的图像时，会出现将唇膏误转移到皮肤上的问题（如图8第一行所示）。而在第二行中，其他方法无法对不同大小的脸部进行转移操作。然而，所提出的AMM模块可以通过计算相似度来准确地为每个像素分配化妆品，从而使我们的结果看起来更好。

4.6. 视频妆容迁移

将视频中的人物妆容进行转移是一项具有挑战性和意义的任务，在实际应用中有着广泛的发展前景。然而，由于视频中人脸的姿态和表情不断变化，这给任务带来了额外的难度。为了检验我们方法的有效性，我们在视频的每一帧上执行了妆容转换，如图9所示。通过采用PSGAN的设计，我们获得了良好且稳定的转换结果。

5. 结论

为了将妆容转移技术应用于实际应用，我们提出了姿态和表情鲁棒的空间感知生成对抗网络（PSGAN），该网络首先将参考妆容风格提炼为两个妆容矩阵，然后

leverages an Attentive Makeup Morphing (AMM) module to conduct makeup transfer accurately. The experiments demonstrate our approach can achieve state-of-the-art transfer results on both frontal facial images and facial images that have various poses and expressions. Also, with the spatial-aware makeup matrices, PSGAN can transfer the makeup partially and adjust the shade of transfer, which greatly broadens the application range of makeup transfer. Moreover, we believe our novel framework can be used in other conditional image synthesis problems that require customizable and precise synthesis.

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利用注意力化妆变形 (AMM) 模块进行准确的化妆转移。实验表明, 我们的方法可以在正面面部图像以及具有各种姿态和表情面部图像上实现最先进的转移效果。此外, 借助空间感知的化妆矩阵, PSGAN可以部分地进行化妆转移并调整转移后的色调, 这大大拓宽了化妆转移的应用范围。我们相信, 我们的新型框架也可以应用于其他需要可定制且精确合成的条件图像合成问题中。

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